

Andrew Gordienko

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Software Verification | Planning Algorithms | Robotics & Hardware Deployment
Seeking a Toronto role | January-April 2027 (4 months)

Education

University of Toronto B.A., Computational Cognitive Science; Computer Science coursework	Expected June 2028 Toronto, Canada
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Technical Skills

Languages: Python, C++, Rust, JavaScript

Tooling: Git, Linux, pytest, Flask, multi-GPU training (H200), Apple MPS

ML & Robotics: PyTorch, MuJoCo, CUDA, NumPy, Gymnasium, Transformers, LeRobot

Work Experience

Automata Technologies <i>Research Intern, Robotics and Planning</i>	May 2026 - Present London, UK
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- Built a deterministic safety gate that verifies search-generated plans before execution, taking the system from simulation to deployment on physical robotic hardware.
- Developed a deadlock search and recovery system using CP-SAT and breadth-first search to identify unsafe workflow states and generate valid recovery paths for medical-laboratory robotics.
- Built a 3D digital twin to generate and validate laboratory configurations, then extended it into a search-based layout optimizer minimizing robot travel and other workflow costs.

OutageHub (OutageHub.ca) <i>Founder</i>	Nov 2024 - Present Hybrid
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- Built a Rust-based platform aggregating Canadian utility outage data into a real-time national map and API.
- Lead partnerships and business development with organizations including the Government of Canada and Canadian Red Cross around outage intelligence and API access.

Research and Projects

TinyVLA	2026
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Vision-Language-Action Model Distillation for Robotics

- Deployed the distilled policy onto physical robotic hardware and debugged the simulation-to-hardware pipeline for closed-loop manipulation.
- Distilled a 450M-parameter SmolVLA policy to 292M parameters for local deployment, reducing bf16 footprint from 749 MB to 557 MB while achieving performance comparable to the teacher on held-out evaluations.

Hudson64	2025
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AlphaZero-Style Reinforcement Learning and Search

- Built an AlphaZero-style chess engine combining a learned policy-value network with Monte Carlo Tree Search, reaching approximately 1800 Elo.
- Implemented PUCT search, network-guided priors, exploration, batched neural evaluation, self-play, replay buffers and arena-based model promotion.
- Scaled training from Apple MPS to NVIDIA H200 GPUs, building high-throughput infrastructure for self-play generation and policy-value training.

UTMIST Virtual Creatures	Sep 2023 - Apr 2024
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Lead Researcher, Reinforcement Learning and Multi-Agent Planning

- Led the creation of a Karl Sims-inspired genetic algorithm to generate MuJoCo robot morphologies and trained shared PPO locomotion policies across evolved bodies to play 2v2 soccer.
- Developed multi-agent planning experiments using Transformer-generated spatial heatmaps and Monte Carlo Tree Search for action selection.

Additional Experience

G&K Software LLC <i>Head of Growth</i>	Nov 2025 - Present Remote
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- Lead sales and business development for a software and AI consultancy, translating customer problems into scoped technical engagements.

CODE-IT Hacks <i>Artificial Intelligence Instructor</i>	Jun 2022 - Sep 2022 Toronto, Canada
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- Taught AI, machine learning and programming to middle and high school students through hands-on coding projects.